



Introduction to Aksha

Advanced AI Solutions for Manufacturing Excellence

Overview of Aksha

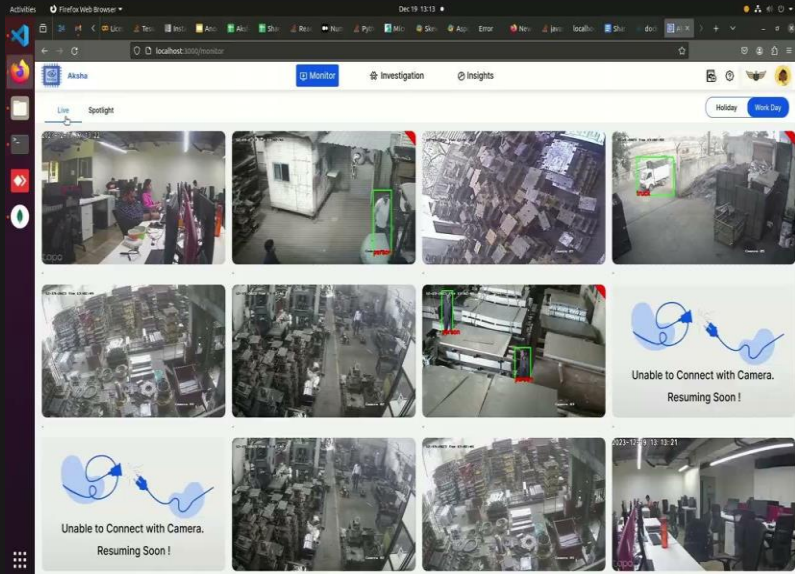


- ▶ Aksha is an intelligent surveillance system designed for automatic monitoring with minimal human intervention.
- ▶ Utilizes advanced technologies such as Computer Vision, Artificial Intelligence, and Generative AI.
- ▶ Key functionalities include object detection, anomaly detection, and real-time analysis.

Architectural Approaches

- ▶ Multiple deployment options: Aksha on Edge, Aksha UI & Backend on Edge, and Aksha on Cloud.
- ▶ Flexible architecture allows for integration with existing systems and scalability.
- ▶ Supports real-time data processing and high availability through Kubernetes and Kafka.

Key Features of Aksha

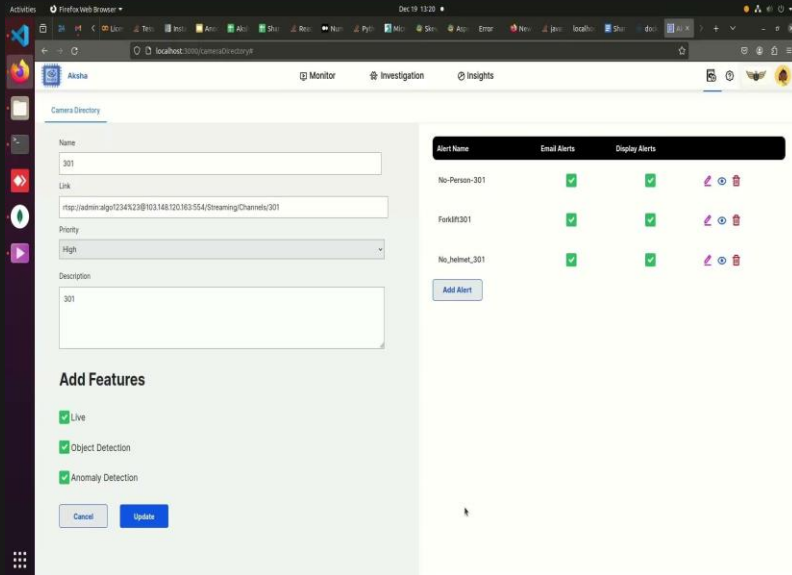


- ▶ Live recording and processing of surveillance data for immediate insights.
- ▶ Advanced object detection capabilities, including identification of personnel, vehicles, and unusual behaviors.
- ▶ Customizable alerts and reporting features, enabling users to tailor notifications to specific needs.

Collaboration and Customization

- ▶ Collaboration with NDIL to customize Aksha for worker safety, enhancing its utility in industrial settings.
- ▶ Implemented a phased approach to development, integrating Generative AI for improved functionality.
- ▶ Successful proof of concept on Raspberry Pi 4B, demonstrating versatility and cost-effectiveness.

Customer Testimonials



- ▶ Clients report significant reductions in human monitoring efforts and enhanced detection of abnormal events.
- ▶ Positive feedback on the system's ability to provide actionable insights and improve safety measures.
- ▶ Customers express eagerness for future updates and innovations from Aksha.

Scalability and Future Prospects

- ▶ Effortless scalability through microservices architecture, enabling easy adaptation to growing needs.
- ▶ Real-time data processing capabilities ensure timely responses to security incidents.
- ▶ Commitment to continuous improvement and innovation in AI-powered surveillance solutions.